

TASK 4: NETWORK INTRUSION DETECTION SYSTEM

Title: Implementation of a Network Intrusion Detection System Using Snort

Introduction

A Network Intrusion Detection System (NIDS) monitors network traffic to identify suspicious activities and potential attacks.

Tools commonly used:

- * Snort
- * Suricata

Step 1: Installation

Install Snort:

```
sudo apt update  
sudo apt install snort
```

Verify installation:

```
snort -V
```

Configure network settings:

```
sudo nano /etc/snort/snort.conf
```

Example:

```
ipvar HOME_NET 192.168.1.0/24
```

Step 2: Configure Rules and Alerts

Create a custom rule:

```
alert icmp any any -> $HOME_NET any (msg:"Ping detected"; sid:1000001; rev:1;)
```

Save:

```
/etc/snort/rules/local.rules
```

Restart:

```
``bash  
sudo systemctl restart snort  
``
```

Step 3: Monitor Network Traffic

Run Snort:

sudo snort -A console -q -c /etc/snort/snort.conf -i eth0

```
Installing dependencies:
  libdaq3  libfastjson4  oinkmaster  snort-common  snort-rules-default
  libestr0 liblognorm5  rsyslog      snort-common-libraries

Suggested packages:
  rsyslog-doc      rsyslog-mongodb      rsyslog-hiredis      rsyslog-docker      | rsyslog-gnutls  snort-doc
  rsyslog-mysql    rsyslog-elasticsearch rsyslog-snmp          rsyslog-clickhouse  rsyslog-gssapi
  rsyslog-pgsql    rsyslog-kafka         rsyslog-kubernetes   rsyslog-openssl     rsyslog-relp

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 0
  Download size: 4,225 kB
  Space needed: 18.2 MB / 15.2 GB available

Ign:1 http://http.kali.org/kali kali-rolling/main amd64 snort-common-libraries amd64 3.12.2.0-0kali1
Ign:2 http://http.kali.org/kali kali-rolling/main amd64 snort-rules-default all 3.12.2.0-0kali1
Ign:3 http://http.kali.org/kali kali-rolling/main amd64 snort-common all 3.12.2.0-0kali1
Get:4 http://http.kali.org/kali kali-rolling/main amd64 libestr0 amd64 0.1.11-2+b2 [9,596 B]
Get:3 http://kali.download/kali kali-rolling/main amd64 snort-common all 3.12.2.0-0kali1 [130 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 snort-rules-default all 3.12.2.0-0kali1 [233 kB]
Get:1 http://kali.download/kali kali-rolling/main amd64 snort-common-libraries amd64 3.12.2.0-0kali1 [291 kB]
Get:5 http://http.kali.org/kali kali-rolling/main amd64 libfastjson4 amd64 1.2304.0-2+b2 [29.2 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 liblognorm5 amd64 2.1.0-1 [72.7 kB]
Get:7 http://kali.download/kali kali-rolling/main amd64 rsyslog amd64 8.2604.0-4 [855 kB]
Get:8 http://kali.download/kali kali-rolling/main amd64 libdaq3 amd64 3.0.24-0kali1 [56.1 kB]
Get:9 http://kali.download/kali kali-rolling/main amd64 snort amd64 3.12.2.0-0kali1 [2,468 kB]
Get:10 http://kali.download/kali kali-rolling/main amd64 oinkmaster all 2.0-5 [79.5 kB]
Fetched 4,225 kB in 18s (236 kB/s)
Selecting previously unselected package snort-common-libraries.
(Reading database... 447175 files and directories currently installed.)
Preparing to unpack .../0-snort-common-libraries_3.12.2.0-0kali1_amd64.deb...
Unpacking snort-common-libraries (3.12.2.0-0kali1)...
Selecting previously unselected package snort-rules-default.
Preparing to unpack .../1-snort-rules-default_3.12.2.0-0kali1_all.deb...
Unpacking snort-rules-default (3.12.2.0-0kali1)...
Selecting previously unselected package snort-common.
Preparing to unpack .../2-snort-common_3.12.2.0-0kali1_all.deb...
Unpacking snort-common (3.12.2.0-0kali1)...
Selecting previously unselected package libestr0:amd64.
Preparing to unpack .../3-libestr0_0.1.11-2+b2_amd64.deb...
Unpacking libestr0:amd64 (0.1.11-2+b2)...
Selecting previously unselected package libfastjson4:amd64.
```

```
(jimoh@kali)-[~]
$ sudo apt install -y snort
The following packages were automatically installed and are no longer required:
  amass-common      libgeos3.14.0      libsnmp40t64      python3-gpg
  curlftpfs         libgirepository-1.0-1  libsodium23      python3-kismetcapturebtgeiger
  enchant-2         libgpgme11t64       libspinxbase3t64  python3-kismetcapturefreaklabszigbee
  girl1.2-girepository-2.0  libgpgmepp6t64      libsqlcipher1     python3-kismetcaptureertl433
  libabsl20240722    libinstpatch-1.0-2  libswscale8       python3-kismetcaptureertladsb
  libann0           libjs-jquery-ui      libteamdctl0       python3-kismetcaptureertlamr
  libarmadillo14     libjs-underscore     libtsk19t64       python3-multipart
  libaudio2         liblab-gamut1        libudfread0        python3-pluginbase
  libavc1394-0       libmbedcrypto16      libunibreak6      python3-protobuf
  libavfilter10     libmjpegutils-2.1-0t64  libvdpau-va-gll   python3-pysmi
  libavformat61     libmongoc-1.0-0t64    libwire shark18    python3-pysnmp4
  libblosc2-4       libmpeg2encpp-2.1-0t64  libwiretap15      python3-terminaltables
  libbluray2        libmpeg2-2.1-0t64     libwsutil16       python3-tz
  libbson-1.0-0t64  libmupdf25.1         libxmlsec1-1       python3-xlrd
  libcdt5           libnet1              libxmlsec1-openssl1  python3-xlutils
  libcgraph6        libplacebo349        libyelp0           python3-xlwt
  libconfig-inifiles-perl  libpocketsphinx3     libzxing3          python3-yaswfp
  libcrypt-dev      libpoppler147        node-cjs-module-lexer  python3-zombie-imp
  libdisplay-info2   libportmidi0         openjdk-21-jre      ruby-unf-ext
  libdlt2           libpostproc58        openjdk-21-jre-headless  samba-ad-ext
  libfuse2t64       libradare2-5.0.0t64  pocketsphinx-en-us  samba-ad-provision
  libgav1-1         libraw23t64          python3-bluepy      samba-dsdb-modules
  libgdal37         librubberband2       python3-click-plugins  vdpau-driver-all
  libgdk-pixbuf2.0-bin  librsync2            python3-fs

Use 'sudo apt autoremove' to remove them.
```

```
Installing:
  snort

Installing dependencies:
  libdaq3  libfastjson4  oinkmaster  snort-common  snort-rules-default
  libestr0 liblognorm5  rsyslog      snort-common-libraries

Suggested packages:
  rsyslog-doc      rsyslog-mongodb      rsyslog-hiredis      rsyslog-docker      | rsyslog-gnutls  snort-doc
  rsyslog-mysql    rsyslog-elasticsearch rsyslog-snmp          rsyslog-clickhouse  rsyslog-gssapi
  rsyslog-pgsql    rsyslog-kafka         rsyslog-kubernetes   rsyslog-openssl     rsyslog-relp
```

```

Selecting previously unselected package snort-rules-default.
Preparing to unpack .../1-snort-rules-default_3.12.2.0-0kali1_all.deb...
Unpacking snort-rules-default (3.12.2.0-0kali1)...
Selecting previously unselected package snort-common.
Preparing to unpack .../2-snort-common_3.12.2.0-0kali1_all.deb...
Unpacking snort-common (3.12.2.0-0kali1)...
Selecting previously unselected package libestr0:amd64.
Preparing to unpack .../3-libestr0_0.1.11-2+b2_amd64.deb...
Unpacking libestr0:amd64 (0.1.11-2+b2)...
Selecting previously unselected package libfastjson4:amd64.
Preparing to unpack .../4-libfastjson4_1.2304.0-2+b2_amd64.deb...
Unpacking libfastjson4:amd64 (1.2304.0-2+b2)...
Selecting previously unselected package liblognorm5:amd64.
Preparing to unpack .../5-liblognorm5_2.1.0-1_amd64.deb...
Unpacking liblognorm5:amd64 (2.1.0-1)...
Selecting previously unselected package rsyslog.
Preparing to unpack .../6-rsyslog_8.2604.0-4_amd64.deb...
Unpacking rsyslog (8.2604.0-4)...
Selecting previously unselected package libdaq3.
Preparing to unpack .../7-libdaq3_3.0.24-0kali1_amd64.deb...
Unpacking libdaq3 (3.0.24-0kali1)...
Selecting previously unselected package snort.
Preparing to unpack .../8-snort_3.12.2.0-0kali1_amd64.deb...
Unpacking snort (3.12.2.0-0kali1)...
Selecting previously unselected package oinkmaster.
Preparing to unpack .../9-oinkmaster_2.0-5_all.deb...
Unpacking oinkmaster (2.0-5)...
Setting up oinkmaster (2.0-5)...
Setting up libestr0:amd64 (0.1.11-2+b2)...
Setting up libfastjson4:amd64 (1.2304.0-2+b2)...
Setting up snort-common (3.12.2.0-0kali1)...
Setting up liblognorm5:amd64 (2.1.0-1)...
Setting up libdaq3 (3.0.24-0kali1)...
Setting up snort-rules-default (3.12.2.0-0kali1)...
Setting up snort-common-libraries (3.12.2.0-0kali1)...
Setting up rsyslog (8.2604.0-4)...
Created symlink '/etc/systemd/system/syslog.service' → '/usr/lib/systemd/system/rsyslog.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/rsyslog.service' → '/usr/lib/systemd/system/rsyslog.service'.
Setting up snort (3.12.2.0-0kali1)...
snort.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.2.6)...
Processing triggers for libc-bin (2.42-16)...

(jimoh@kali)-[~]
$

```

The system continuously monitors traffic and displays alerts.

Step 4: Response Mechanisms

Responses include:

- * Alert notifications
- * Logging incidents
- * Blocking suspicious IPs

Example:

```
sudo iptables -A INPUT -s 192.168.1.100 -j DROP
```

Step 5: Visualization Dashboard

Tools:

- * Kibana
- * Grafana

Dashboard features:

- * Attack trends
- * Alert graphs
- * Traffic monitoring
- * Threat statistics

Conclusion

The NIDS detects network threats early and improves overall system security.